

Making up infant formula at a Target concentration

Full working for every exercise. Numbers may differ in the last decimal place if you rounded at a different point, but the recipe should match.

1 Reading a label

Work out the displacement and kcal per scoop of Formula A.

Formula A: the label says

Energy: 68 kcal per 100 mL

Reconstitution statement: 2 scoops + 60 mL water makes approximately 67 mL

Grams per scoop: 1 scoop = 4.6 g

	Step	Working	Answer
A1	Scoops in the statement	Given on the label	2 scoops
A2	Displacement (mL per g)	$(67 - 60) \div (2 \times 4.6)$	0.761 mL per g
A3	kcal per scoop	$68 \div 100 \times (67 \div 2)$	22.8 kcal

2 Reading a label

Work out the displacement and kcal per scoop of Formula B.

Formula B: the label says

Energy: 66 kcal per 100 mL

Reconstitution statement: 132 g powder + 900 mL water makes 1 L

Grams per scoop: Average scoop weight = 4.4 g

	Step	Working	Answer
A1	Scoops in the statement	$132 \div 4.4$	30 scoops
A2	Displacement (mL per g)	$(1000 - 900) \div 132$	0.758 mL per g
A3	kcal per scoop	$66 \div 100 \times (1000 \div 30)$	22.0 kcal

3 Building a recipe

Formula C has **21.5 kcal per scoop**, **4.5 g per scoop** and a displacement of **0.711 mL per g**. Make **180 mL** at **20 kcal/30 mL**.

Step	Working	Answer
1 kcal/mL	$20 \div 30$	0.667
2 kcal needed	180×0.667	120 kcal
3 Scoops	$120 \div 21.5 = 5.58$, round up	6 scoops
4 Actual kcal	6×21.5	129.0 kcal
5 Total powder (g)	6×4.5	27.0 g
6 Total displacement (mL)	27.0×0.711	19.2 mL
7 Water (mL)	$129.0 \div 0.667 - 19.2 = 174.3$, round up to 5 mL	175 mL
8 Actual volume (mL)	$175 + 19.2$	194.2 mL
9 Actual concentration (kcal/30 mL)	$129.0 \div 194.2 \times 30$	19.93 kcal/30 mL

Recipe: 6 scoops + 175 mL water

Makes: 194.2 mL at 19.93 kcal/30 mL

0.5 kcal/30 mL or more under target? No

4 Building a recipe

Formula C has **21.5 kcal per scoop**, **4.5 g per scoop** and a displacement of **0.711 mL per g**. Make **150 mL** at **27 kcal/30 mL**.

Step	Working	Answer
1 kcal/mL	$27 \div 30$	0.9
2 kcal needed	150×0.9	135 kcal
3 Scoops	$135 \div 21.5 = 6.28$, round up	7 scoops
4 Actual kcal	7×21.5	150.5 kcal
5 Total powder (g)	7×4.5	31.5 g
6 Total displacement (mL)	31.5×0.711	22.4 mL
7 Water (mL)	$150.5 \div 0.9 - 22.4 = 144.8$, round up to 5 mL	145 mL
8 Actual volume (mL)	$145 + 22.4$	167.4 mL
9 Actual concentration (kcal/30 mL)	$150.5 \div 167.4 \times 30$	26.97 kcal/30 mL

Recipe: 7 scoops + 145 mL water

Makes: 167.4 mL at 26.97 kcal/30 mL

0.5 kcal/30 mL or more under target? No

5 Building a recipe

Formula D has **36 kcal per scoop**, **7 g per scoop** and a displacement of **0.714 mL per g**. Make **60 mL** at **30 kcal/30 mL**.

Step	Working	Answer
1 kcal/mL	$30 \div 30$	1
2 kcal needed	60×1	60 kcal
3 Scoops	$60 \div 36 = 1.67$, round up	2 scoops
4 Actual kcal	2×36	72.0 kcal
5 Total powder (g)	2×7	14.0 g
6 Total displacement (mL)	14.0×0.714	10.0 mL
7 Water (mL)	$72.0 \div 1 - 10.0 = 62.0$, round up to 5 mL	65 mL
8 Actual volume (mL)	$65 + 10.0$	75.0 mL
9 Actual concentration (kcal/30 mL)	$72.0 \div 75.0 \times 30$	28.80 kcal/30 mL

Recipe: 2 scoops + 65 mL water

Makes: 75.0 mL at 28.80 kcal/30 mL

0.5 kcal/30 mL or more under target? Yes: double-check before using

6 From label to bottle

Make **180 mL** at **24 kcal/30 mL** of Formula E.

Formula E: the label says

Energy: 67 kcal per 100 mL

Reconstitution statement: 135 g powder + 900 mL water makes 1 L

Grams per scoop: Average scoop weight = 4.5 g

Part A · Reading the label

	Step	Working	Answer
A1	Scoops in the statement	$135 \div 4.5$	30 scoops
A2	Displacement (mL per g)	$(1000 - 900) \div 135$	0.741 mL per g
A3	kcal per scoop	$67 \div 100 \times (1000 \div 30)$	22.3 kcal

Part B · Building the recipe

	Step	Working	Answer
1	kcal/mL	$24 \div 30$	0.8
2	kcal needed	180×0.8	144 kcal
3	Scoops	$144 \div 22.3 = 6.46$, round up	7 scoops
4	Actual kcal	7×22.3	156.1 kcal
5	Total powder (g)	7×4.5	31.5 g
6	Total displacement (mL)	31.5×0.741	23.3 mL
7	Water (mL)	$156.1 \div 0.8 - 23.3 = 171.8$, round up to 5 mL	175 mL
8	Actual volume (mL)	$175 + 23.3$	198.3 mL
9	Actual concentration (kcal/30 mL)	$156.1 \div 198.3 \times 30$	23.61 kcal/30 mL

Recipe: 7 scoops + 175 mL water

Makes: 198.3 mL at 23.61 kcal/30 mL

0.5 kcal/30 mL or more under target? No

7 From label to bottle

Make 120 mL at 27 kcal/30 mL of Formula F.

Formula F: the label says

Energy: 70 kcal per 100 mL

Reconstitution statement: 1 scoop + 30 mL water makes approximately 33 mL

Grams per scoop: 1 scoop = 4.4 g

Part A · Reading the label

	Step	Working	Answer
A1	Scoops in the statement	Given on the label	1 scoop
A2	Displacement (mL per g)	$(33 - 30) \div 4.4$	0.682 mL per g
A3	kcal per scoop	$70 \div 100 \times 33$	23.1 kcal

Part B · Building the recipe

	Step	Working	Answer
1	kcal/mL	$27 \div 30$	0.9
2	kcal needed	120×0.9	108 kcal
3	Scoops	$108 \div 23.1 = 4.68$, round up	5 scoops
4	Actual kcal	5×23.1	115.5 kcal
5	Total powder (g)	5×4.4	22.0 g
6	Total displacement (mL)	22.0×0.682	15.0 mL
7	Water (mL)	$115.5 \div 0.9 - 15.0 = 113.3$, round up to 5 mL	115 mL
8	Actual volume (mL)	$115 + 15.0$	130.0 mL
9	Actual concentration (kcal/30 mL)	$115.5 \div 130.0 \times 30$	26.65 kcal/30 mL

Recipe: 5 scoops + 115 mL water

Makes: 130.0 mL at 26.65 kcal/30 mL

0.5 kcal/30 mL or more under target? No

8 Explain

1. Why isn't the water simply the Target volume?

The powder takes up room too. Every gram of it adds its displacement to the volume, so water equal to the Target volume would make more than the Target volume, at a lower concentration than intended.

2. Why are scoops rounded up rather than to the nearest whole scoop?

You can only use whole scoops, and rounding up makes sure there is at least enough energy. The water is then sized to dilute that energy back to the Target concentration. Rounding down would leave too little energy for the Target volume.

3. A recipe's Actual concentration comes out 0.5 kcal/30 mL or more below the Target concentration. What should you do?

Double-check the inputs and the working before using it. A large drop usually means a small Target volume with a large scoop, where one scoop is a big step; confirm the result is acceptable for the patient.